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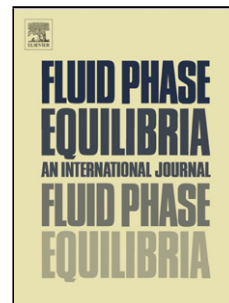
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# Fully Implicit Compositional Simulator for Modeling of Asphaltene Deposition during Natural Depletion

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## Abstract

This paper describes development of a fully implicit compositional simulator for modeling of asphaltene deposition during natural depletion. In this paper, a new approach for multiphase flash calculation has been developed. This approach provides a more detailed description of the kinetic part of asphaltene deposition which needs to be explained more clearly. Due to a large number of unknowns, there are many ways to solve such a system by choosing different sets of independent variables. A new set of independent variables in a fully implicit model is considered for asphaltene deposition modeling. By incorporating of an asphaltene precipitation model into a compositional simulator where the phase equilibrium equations, the volumetric constraint equation, the component transport equations, the multiphase flash equations and the deposition equation are solved simultaneously, a simulator for asphaltene deposition was developed with respect to natural depletion. The pure solid model is used to model asphaltene precipitation. The solid particles are considered to be separated into three parts: precipitated, flocculated and deposited solid. A first-order chemical reaction is used which models forward and reverse rates for the conversion of precipitated asphaltene to flocculated asphaltene. Also, a deposition model including adsorption, pore throat plugging, and re-entrainment was used. The simulator can also predict formation damage including porosity and permeability reduction in each block.

**Keywords:** Asphaltene, modeling, compositional simulator, solid model, deposition

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## 1. Introduction

Asphaltene deposition phenomena in the oil reservoir during natural depletion is a complex and destructive problem which the most of the oil field reservoir, production and/or process facilities is encountered with. Wettability alterations, relative permeability reduction, blockage of the flow with additional pressure drop in wellbore tubing, upstream process facilities and surface pipelines are the common problems which created from asphaltene deposition[1]. By having a good prediction of location and amount of asphaltene precipitation and deposition, oil field operators will be able to design an efficient operation process to avoid the risks associated with asphaltene deposition [2].

Various thermodynamic models have been presented to model the phase behavior of asphaltene precipitation. The liquid solubility models, the pure solid models, the colloidal solution models and a thermodynamic micellization model are the four main groups of thermodynamic models which are available in the literature.

The liquid solubility models are based on polymer solution theories by Flory-Huggins [3]. The first of such a model was developed by Hirshberg et al. [4]. According to this model, asphaltene stability is described in terms of reversible solution equilibrium.

In the solid models, EOS is applied to model oil and gas phases and asphaltene is considered as a single dense phase. Nghiem [5] and Chung [6] applied the solid model approach to simulate phase behavior of asphaltene precipitation.

In 1987, Leontaritis and Mansoori [7] established a thermodynamic colloidal model to simulate phase behavior of asphaltene precipitation. This method assumes that asphaltene can be found in the oil as colloidal particles peptized by resins.

The thermodynamic micellization model was initially developed by Pan and Firoozabadi [8, 9]. It assumes that asphaltene micelle is stabilized by resin molecules. Minimization of the

total Gibbs energy is the basis of this approach to determine the structure and concentration of the micelle [10].

Even though none of these models satisfactorily match the experimental data, some are simple and fast and have been implemented in reservoir simulators [11, 12]. The solid models have been widely used in the reservoir simulators due to the flexibility and simple application [13]. In addition, the main aim of this research is to develop an EOS compositional

simulator. Thus, a thermodynamic model must be used to be compatible with flash calculation in an EOS simulator. Considering these reasons, the solid model is used for the modeling of asphaltene precipitation.

The pure solid model has been enhanced by allowing the precipitated asphaltene to be converted to larger flocculated particles according to a simple chemical reaction. The use of forward and backward reaction rates allows this conversion to be specified as fully irreversible, fully reversible or partially reversible [14].

A few models have been proposed to describe asphaltene deposition. Leontaritis [15] presented the first model to predict asphaltene deposition in the near wellbore region with the assumption of constant production rate. It was assumed that asphaltene deposition occurs only around the wellbore vicinity and the area of formation damage caused by asphaltene deposition was constant [16]. Nghiem et al.'s model [17] assumes that the asphaltene deposition is a pure adsorption process. The surface deposition, pore throat plugging and deposit entrainment were observed in asphaltene deposition rather than adsorption [18]. Nghiem's model was criticized for ignoring deposition in the pore space which was considered to be an important factor. The surface deposition, pore throat plugging and entrainment were presented in Wang and Civan's deposition model [19]. It was assumed that asphaltene can deposit everywhere rather than the near wellbore region.

In this work pure solid model is used to model asphaltene precipitation[20, 21] . A first-order chemical reaction was employed to model forward and reverse rates for the conversion of precipitated asphaltene to flocculated asphaltene. Also, a deposition model including adsorption, pore throat plugging, and re-entrainment is used. The reduction in the rock porosity and permeability are also included in the asphaltene model[22].

This paper talks about developing new method to calculate asphaltene precipitation through thermodynamics point of view. Also, the details of dynamic aspect of asphaltene precipitation which needs to be explained more clearly are described. The implementation of an asphaltene thermodynamic model into a fully implicit, multiphase, compositional simulator was the challenging part of this research. A new set of independent unknowns in a fully implicit scheme was presented for asphaltene deposition modeling. In order to find the solution of these variables, the same number of equations was also presented. The mathematical formulation for the derivatives of the governing equations for building the Jacobian matrix and the residuals of the governing equations are described. In addition, the techniques used to implement formulations in Matlab code are described. The asphaltene precipitation model in Matlab code was compared to CMG's compositional simulator, called GEM.

## **2. Asphaltene precipitation model**

The asphaltene precipitation, flocculation and deposition models used in the simulator are discussed in detail in this paper.  $S_1$ ,  $S_2$  and  $S_3$  represent asphaltene precipitation, flocculation and deposition respectively. Kohse and Nghiem (2004) proposed a model which assumes the heaviest component of oil can be splitted into a non-precipitating and precipitating component. The precipitated solid is divided into solid 1 which is in equilibrium with the heaviest component in the oil phase and solid 2 that is created from solid1 via a first order chemical reaction. Solid 2 can be used to represent the flocculation of smaller precipitated

asphaltene particles into larger aggregates. This process is modeled by a set of two kinetic reactions to allow reversibility between aggregation of the fines into flocs and dissociation of the flocs into fines. Once the asphaltene component become flocs they are available to deposit onto the rock. Solid 1 flows as suspended particles in the oil phase, while Solid 2 may flow with the oil or deposit in the rock matrix [14]. That is, Solid 2 (flocculated asphaltene) can be converted to S1 (precipitated asphaltene) through the reverse reaction or converted to Solid 3 (deposited asphaltene) via the deposition rate equation or stay in the state of flocs. The conversion of solid 2 to solid 3 is modeled by Wang's model (2001). In the implementation of the deposition model, only the flocculated particles (solid S2) are considered to deposit.

When the oil, gas, and solid phases coexist, the following thermodynamic equilibrium equations are satisfied:

$$\ln f_{io} = \ln f_{ig}, i = 1, \dots, n_c \quad (1)$$

$$\ln f_{n_c,o} = \ln f_{sl} \quad (2)$$

Equation (1) shows the equality of fugacity component  $i$  in the oil and in the gas phase, and Equation (2) shows the equality of fugacity of the solid  $S_1$  and the precipitating component in the oil phase. The solid fugacity is calculated from:

$$\ln f_{sl} = \ln f_{sl}^* + \frac{V_s(P - P^*)}{RT} \quad (3)$$

Where  $f_{s_1}$  is the fugacity of solid  $S_1$  at reservoir pressure,  $f_{sl}^*$  is the reference solid fugacity at  $P^*$ ,  $V_s$  is the solid molar volume and  $P^*$  is the pressure at which the asphaltene just starts to precipitate. The reference pressure is obtained from experimental data.

## 2.1 Specification of solid molar volume

The solid molar volume is set as a matching parameter in the asphaltene precipitation model. The molar volume of the precipitating component which is set as an initial value is predicted by the equation of state at the reference pressure ( $P^*$ ). The Peng-Robinson equation-of-state was used for all calculation in this research. This equation improves the liquid density prediction and the PR-EOS is perhaps the most popular and widely used EOS for the hydrocarbon phase behavior calculations. To find a desired quantity of solid molar volume, the error term versus solid molar volume is plotted. The value of solid molar volume which gives minimum error is fixed for all subsequent asphaltene precipitation calculations. The error can be written as follows:

$$\text{error} = \sum_{i=1}^n \frac{|(W_{(i)\text{exp}} - W_{(i)\text{model}})|}{|W_{(i)\text{exp}}|} \quad (4)$$

**Figure 1** shows the relative error as a function of solid molar volume. The experimental data ( $W_{\text{exp}}$ ) which is taken from Burke et al. [23] is shown in **Table 1**. The EOS calculated for the solid molar volume of precipitating component is 10.5611 ft<sup>3</sup>/lb<sub>m</sub>. As can be seen, minimum error has occurred when the solid molar volume is given as 10.81215 ft<sup>3</sup>/lb<sub>m</sub>.

## 2.2 Flash calculation

Flash calculation is performed to determine the phase fraction and compositions of each phase. In multiphase flash calculation, the number of phases is generally unknown beforehand. The phase stability analysis is used to determine the number of phases in each grid block.

To test the existence of the solid phase, the following criteria should be satisfied. The solid phase exists if

$$\ln f_{n_o} \geq \ln f_{s1} \quad (5)$$

**2.2.1 Multiphase Flash Calculations.** Considering that the flocculation and deposition processes change the fluid composition, the following equations apply:

$$N_i = N_{i0} + N_{ig} \quad , i = 1, \dots, n_c - 1 \quad (6)$$

$$N_{nc} = N_{nc0} + N_{ncg} + N_{s1} + N_{s2} + N_{s3} \quad (7)$$

$$N'_{nc} = N_{nc0} + N_{ncg} + N_{s1} \quad (8)$$

$$N_T = \sum_{i=1}^{n_c} N_i \quad (9)$$

$$N'_T = N_T - N_{s2} - N_{s3} \quad (10)$$

$$z_i = \frac{N_i}{N'_T} \quad , i = 1, \dots, n_c - 1 \quad (11)$$

$$z_{nc} = \frac{N'_{nc}}{N'_T} \quad (12)$$

Where  $N_{s1}$  represents the moles of precipitated asphaltene per pore volume,  $N_{s2}$  is the moles of flocculated asphaltene per pore volume and  $N_{s3}$  is the moles of deposited asphaltene per pore volume.

The equilibrium equations can be written in terms of K-values as:

$$\ln K_{ig} + \ln \phi_{ig} - \ln \phi_{i0} = 0 \quad (13)$$



$$\ln K_{n_c s} + \ln \varphi_s - \ln \varphi_{n_c o} = 0 \quad (14)$$

With

$$K_{ig} = \frac{y_{ig}}{y_{io}} \quad (15)$$

$$\varphi_{ij} = \frac{f_{ij}}{y_{ij} P} \quad (16)$$

$j=o, g$

$$K_{n_c s} = \frac{1}{y_{n_c o}} \quad (17)$$

$$\varphi_s = \frac{f_s}{P} \quad (18)$$

In conjunction with the above equations the following material balance equations can be derived:

$$\sum_{i=1}^{nc} \frac{(K_{ig} - 1) z_i}{1 + L_g (K_{ig} - 1) + L_s (K_{is} - 1)} = 0 \quad (19)$$

$$\sum_{i=1}^{nc} \frac{(K_{is} - 1) z_i}{1 + L_g (K_{ig} - 1) + L_s (K_{is} - 1)} = 0 \quad (20)$$

The oil and gas phase composition can be calculated from the following equations:

$$y_{io} = \frac{z_i}{1 + L_g (K_{ig} - 1) + L_s (K_{is} - 1)} \quad (21)$$

$$y_{ig} = K_{ig} x_{io} \quad (22)$$

### 3. Mathematical Model Description

Independent unknowns are chosen as,  $N_i$ ,  $\ln(K_i)$ ,  $N_{s2}$ ,  $N_{s3}$ ,  $P$ ,  $K_s$ ,  $L_g$ ,  $L_s$  which gives  $2n_c + 6$  independent variables for each grid block. Note that  $P$  is the oil phase pressure,  $N$  is moles per unit pore volume and  $L_s$  is the mole fraction of precipitated asphaltene. We need the same number of equations to be able to find the solution of these variables. In order to model the deposition of asphaltene, the following set of equations is presented as follows:

**3.1 Mass conservation equations.** The general mass conservation equation with the consideration of asphaltene for species  $j$  ( $j = 1, \dots, n_c$ ) can be expressed as:

$$-\nabla(\xi_o y_{jo} u_o) - \nabla(\xi_g y_{jg} u_g) + q_j = \frac{\partial(\phi N_j)}{\partial t} \quad j = 1, \dots, n_c - 1 \quad (23)$$

$$-\nabla(\xi_o (y_{n_c o} + y_{s1} + y_{s2}) u_o) - \nabla(\xi_g y_{n_c g} u_g) + q_{n_c} = \frac{\partial(\phi N_{n_c})}{\partial t} \quad (24)$$

Where,

$$y_{s1} = \frac{N'_T L_s}{N_o} \quad (25)$$

$$y_{s2} = \frac{N_{s2}}{N_o} \quad (26)$$

In porous media, the phase velocity  $u_{o,g}$  corresponds to the Darcy velocity:

$$u_{o,g} = -\lambda_{o,g} \nabla \Phi_{o,g} \quad (27)$$

$$\lambda_{o,g} = \frac{KK_{ro,g}}{\mu_{o,g}} \quad (28)$$

Substituting equation (27) into equations (23) and (24) yields:

$$\nabla(\xi_o y_{jo} \lambda_o \nabla \Phi_o) + \nabla(\xi_g y_{jg} \lambda_g \nabla \Phi_g) + q_j = \frac{\partial(\phi N_j)}{\partial t} \quad j=1, \dots, n_c - 1 \quad (29)$$

$$\nabla(\xi_o (y_{n_c o} + y_{s_1} + y_{s_2}) \lambda_o \nabla \Phi_o) + \nabla(\xi_g y_{n_c g} \lambda_g \nabla \Phi_g) + q_{n_c} = \frac{\partial(\phi N_{n_c})}{\partial t} \quad (30)$$

The finite-difference form of the material balance equation is presented as follows:

$$R(i, k) = \Delta T_{io}^{n+1} y_{(i,j),o}^{n+1} (\Delta P_i^{n+1} - \gamma_{io}^{n+1} \Delta d) + \Delta T_{ig}^{n+1} y_{(i,j),g}^{n+1} (\Delta P_i^{n+1} + \Delta P_{icog}^{n+1} - \gamma_{ig}^{n+1} \Delta d) + q_{io}^{n+1} \xi_{io}^{n+1} (y_{(i,j),o})^{n+1} + q_{ig}^{n+1} \xi_{ig}^{n+1} (y_{(i,j),g})^{n+1} - \frac{V_i}{\Delta t} \left( (\phi_i N_{(i,j)})^{n+1} - (\phi_i N_{ij})^n \right) \quad i = 1, \dots, n_b, \quad k, j = 1, \dots, n_c - 1 \quad (31)$$

Where  $n_b$  represents the total number of grid cells and  $n_c$  is the number of hydrocarbon components.

$$R(i, n_c) = \Delta T_{io}^{n+1} (y_{(i,n_c),o} + y_{is1} + y_{is2})^{n+1} (\Delta P_i^{n+1} - \gamma_{io}^{n+1} \Delta d) + \Delta T_{ig}^{n+1} (y_{(i,n_c),g})^{n+1} (\Delta P_i^{n+1} + \Delta P_{icog}^{n+1} - \gamma_{ig}^{n+1} \Delta d) + q_{io}^{n+1} \xi_{io}^{n+1} (y_{(i,n_c),o} + y_{is1} + y_{is2})^{n+1} + q_{ig}^{n+1} \xi_{ig}^{n+1} (y_{(i,n_c),g})^{n+1} - \frac{V_i}{\Delta t} \left( (\phi_i N_{(i,n_c)})^{n+1} - (\phi_i N_{(i,n_c)})^n \right) \quad (32)$$

Where,

$$T_{o,g} = \frac{KA}{\Delta l} \left( \frac{K_{ro,g} \xi_{o,g}}{\mu_{o,g}} \right) \quad (33)$$

**3.2 Phase Equilibrium Equations.** When the oil, gas, and solid phases coexist, the following thermodynamic equilibrium equations are satisfied:

$$R(i, k) = \ln K_{(i,j)g} + \ln \phi_{(i,j)g} - \ln \phi_{(i,j)o}$$

$$j=1, \dots, n_c, \quad k=n_c+1, \dots, 2n_c \quad (34)$$

$$R(i, 2n_c+1) = \ln K_{is} + \ln \phi_{is} - \ln \phi_{(i,n_c)o} \quad (35)$$

**3.3 Multiphase Flash Equations.** Multiphase flash equations can be show as:

$$R(i, 2n_c+2) = \sum_{j=1}^{n_c} \frac{(K_{(i,j)g} - 1)z_{(i,j)}}{1 + L_{ig}(K_{(i,j)g} - 1) + L_{is}(K_{is} - 1)} \quad (36)$$

$$R(i, 2n_c+3) = \sum_{j=1}^{n_c} \frac{(K_{is} - 1)z_{(i,j)}}{1 + L_{ig}(K_{(i,j)g} - 1) + L_{is}(K_{is} - 1)} \quad (37)$$

**3.4 Volumetric Constraint Equation.** The summation of volumes of each volume-occupying component should be the same as pore volume. The constraint can be expressed as follows:

$$R(i, 2n_c+4) = \frac{N_{io}}{\xi_{io}} + \frac{N_{ig}}{\xi_{ig}} + \frac{N_{is}}{\xi_{is}} - 1 \quad (38)$$

**3.5 Mass conservation equation for solid  $s_2$ [14].** A material balance on solid  $s_2$  gives:

$$R(i, 2n_c+5) = \Delta T_{io}^{n+1} (y_{is2})^{n+1} (\Delta P_i^{n+1} - \gamma_{io}^{n+1} \Delta d) + q_{io}^{n+1} \xi_{io}^{n+1} (y_{is2})^{n+1} + V\phi S_o r$$

$$- \frac{V}{\Delta t} ((\phi_1 N_{is2})^{n+1} - (\phi_1 N_{is2})^n) \quad (39)$$

$$r = K_{12} C_{s1,o} - K_{21} C_{s2,o} \quad (40)$$

Where,

$$C_{s1,o} = \frac{N'_T L_s}{S_o} \quad (41)$$

$$C_{s1,o} = \frac{N_{s2}}{S_o} \quad (42)$$

### 3.6 Asphaltene Deposition Equation

In this model, only solid  $s_2$  is considered to deposit. Physically, this implies that the small asphaltene precipitate particles are more likely to flow with the oil phase, while the larger aggregates are more likely to deposit on the reservoir rock [14]. The flocs can adsorb on the rock surface, can be trapped within the porous media because of their size (plugging) or can be entrained and returned to the oil phase because of high, local velocity (shear) [24]. Wang and Civan (2001) introduced the following equation for asphaltene deposition rate [25]:

$$\frac{\partial V_{(s_2^d)}}{\partial t} = \alpha C_{(s_2^f)} \phi - \beta V_{(s_2^d)} (V_o - V(cr, o)) + \gamma u_o C_{(s_2^f)} \quad (43)$$

The discretized form of the deposition rate equation is:

$$R(i, 2n_c + 6) = (V_{(s_2^d)}^{(n+1)} - V_{(s_2^d)}^n) / \Delta t - \alpha C_{(s_2^f)}^{(n+1)} \phi^{(n+1)} + \beta V_{(s_2^d)}^{(n+1)} (V_o^n - V(cr, o)) - \gamma u_o^n C_{(s_2^f)}^{(n+1)} \quad (44)$$

**3.7 Porosity and Permeability Reduction.** A power law relationship can be used to calculate the resistance factor [22, 25]. We use the following equation to update the permeability values.

$$R_f = \frac{K_0}{K} = \left( \frac{\phi_0}{\phi} \right)^3 \quad (45)$$

The instantaneous local porosity is equal to the initial porosity ( $\phi_0$ ) minus the amount of deposited solid:

$$\phi = \phi_0 - V_{s_2^d} \quad (46)$$

#### 4. Solution Approach

we solved the presented equations by constructing a Jacobian matrix,  $J$ , and a residual vector,  $R$ , with independent variables:  $N_i$ ,  $\ln(K_i)$ ,  $N_{s_2}$ ,  $N_{s_3}$ ,  $P$ ,  $K_s$ ,  $L_g$ ,  $L_s$ . A fully implicit technique is used to solve the governing equation for these independent variables.

There are  $(2n_c + 6)n_b$  nonlinear equations for the same number of independent variables.

This system can be written as follows:

$$\vec{X} = (\vec{X}_1, \vec{X}_2, \dots, \vec{X}_{n_b}) \quad (47)$$

$$\vec{R} = (\vec{R}_1, \vec{R}_2, \dots, \vec{R}_{n_b}) \quad (48)$$

The unknown vector for  $i^{th}$  gridblock is:

$$\vec{X}_i = (x(i,1), \dots, x(i, 2n_c + 6))^T \quad (49)$$

Where

$$x(i, j) = N(i, j)$$

$$i = 1, \dots, n_b, \quad j = 1, \dots, n_c \quad (50)$$

$$x(i, j) = \ln K(i, j) \quad i = 1, \dots, n_b, \quad j = n_c + 1, \dots, 2n_c \quad (51)$$

And we set:

$$x(i, 2n_c + 1) = N_{is3} \quad (52)$$

$$x(i, 2n_c + 2) = N_{is2} \quad (53)$$

$$x(i, 2n_c + 3) = P_i \quad (54)$$

$$x(i, 2n_c + 4) = K_{is} \quad (55)$$

$$x(i, 2n_c + 5) = L_{ig} \quad (56)$$

$$x(i, 2n_c + 6) = L_{is} \quad (57)$$

And the residual vector for  $i^{th}$  gridblock is:

$$\vec{R}_i = (R(i,1), \dots, R(i, 2n_c + 6))^T \quad (58)$$

Matlab contains routines which can be used to solve systems of non-linear equation. We will now use fsolve to achieve the solution. First, fsolve requires an initial guess, which usually uses the solution vector at old timestep as the initial guess for the new timestep. The second step is the calculation of the residual vector from initial guesses of the independent variables. To calculate residual vector, it is necessary to evaluate dependent variables. In the third step, check the convergence criteria. If the convergence is achieved, it outputs the results at the current time level and then returns to the framework for the next timestep. If the convergence failed, the Jacobian matrix (J) and the residual vector are updated and Steps 2 and 3 are then repeated. **Fig. 2** shows a flow chart for this process. Independent variables are updated as follows:

$$\vec{X}^{new} = \vec{X}^{old} + \Delta \vec{X} \quad (59)$$

Where  $\Delta \vec{X}$  is calculate from  $\left(\vec{X}^{old}\right) \Delta \vec{X} = -\vec{R}\left(\vec{X}^{old}\right)$ . This equation can be rewritten in matrix form:

$$\begin{bmatrix} J_{1,1} & J_{1,2} & \cdots & J_{1,n_b} \\ J_{2,1} & J_{2,2} & \cdots & J_{2,n_b} \\ \vdots & \vdots & \ddots & \vdots \\ J_{n_b,1} & J_{n_b,2} & \cdots & J_{n_b,n_b} \end{bmatrix} \begin{bmatrix} \Delta \vec{X}_1 \\ \Delta \vec{X}_2 \\ \vdots \\ \Delta \vec{X}_{n_b} \end{bmatrix} = - \begin{bmatrix} \vec{R}_1 \\ \vec{R}_2 \\ \vdots \\ \vec{R}_{n_b} \end{bmatrix} \quad (60)$$

Where  $J_{I,J}$  is the Jacobian matrix for  $I^{th}$  gridcell with respect to unknowns of the  $j^{th}$  grid cell. The general structure of the jacobian matrix is shown as follows:

$$J_{I,J} = \begin{bmatrix} \frac{\partial R(I,1)}{\partial x(J,1)} & \frac{\partial R(I,1)}{\partial x(J,2)} & \cdots & \frac{\partial R(I,1)}{\partial x(J,2n_c+6)} \\ \frac{\partial R(I,2)}{\partial x(J,1)} & \frac{\partial R(I,2)}{\partial x(J,2)} & \cdots & \frac{\partial R(I,2)}{\partial x(J,2n_c+6)} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial R(I,2n_c+6)}{\partial x(J,1)} & \frac{\partial R(I,2n_c+6)}{\partial x(J,2)} & \cdots & \frac{\partial R(I,2n_c+6)}{\partial x(J,2n_c+6)} \end{bmatrix}$$

Note that only mass conservation equations involve variables from adjacent gridblocks, as a result, the corresponding rows of the Jacobian matrix of the other equations do not contain off-diagonal elements [26].

There are special cases during the simulation time that make us to create some modification in the residual equations. For example, if only one hydrocarbon phase is exist, the derivatives with respect to  $\ln K_i, N_{s2}, N_{s3}, K_s, L_g$  and  $L_s$  are not needed in the Jacobian matrix. If we set the corresponding residual equation of these variables to zero, all the elements of corresponding rows of the Jacobian built from Matlab for these equations will be equal to zero. To overcome this problem, the residual equations corresponding to these variables need to be manipulated. The Jacobian matrix for this case is as follows:



|                                                |          |                                                  |                        |                                                     |                                                     |                                                     |                                                     |          |                                                     |
|------------------------------------------------|----------|--------------------------------------------------|------------------------|-----------------------------------------------------|-----------------------------------------------------|-----------------------------------------------------|-----------------------------------------------------|----------|-----------------------------------------------------|
| $\frac{\partial R(I,1)}{\partial x(J,1)}$      | ...      | $\frac{\partial R(I,1)}{\partial x(J,n_c)}$      | 0 ... 0                | $\frac{\partial R(I,1)}{\partial x(J,2n_c+1)}$      | $\frac{\partial R(I,1)}{\partial x(J,2n_c+2)}$      | $\frac{\partial R(I,1)}{\partial x(J,2n_c+3)}$      | $\frac{\partial R(I,1)}{\partial x(J,2n_c+4)}$      | 0        | $\frac{\partial R(I,1)}{\partial x(J,2n_c+6)}$      |
| $\vdots$                                       | $\vdots$ | $\vdots$                                         | $\vdots \vdots \vdots$ | $\vdots$                                            | $\vdots$                                            | $\vdots$                                            | $\vdots \vdots$                                     | $\vdots$ | $\vdots$                                            |
| $\frac{\partial R(I,n_c)}{\partial x(J,1)}$    | ...      | $\frac{\partial R(I,n_c)}{\partial x(J,n_c)}$    | 0 ... 0                | $\frac{\partial R(I,n_c)}{\partial x(J,2n_c+1)}$    | $\frac{\partial R(I,n_c)}{\partial x(J,2n_c+2)}$    | $\frac{\partial R(I,n_c)}{\partial x(J,2n_c+3)}$    | $\frac{\partial R(I,n_c)}{\partial x(J,2n_c+4)}$    | 0        | $\frac{\partial R(I,n_c)}{\partial x(J,2n_c+6)}$    |
| 0                                              | ...      | 0                                                | 1 ... 0                | 0                                                   | 0                                                   | 0                                                   | 0 0                                                 | 0        | 0                                                   |
| $\vdots$                                       | $\vdots$ | $\vdots$                                         | $\vdots \vdots \vdots$ | $\vdots$                                            | $\vdots$                                            | $\vdots$                                            | $\vdots \vdots$                                     | $\vdots$ | $\vdots$                                            |
| 0                                              | ...      | 0                                                | 0 ... 0                | 0                                                   | 0                                                   | 0                                                   | 0 0                                                 | 0        | 0                                                   |
| $\frac{\partial R(I,2n_c+1)}{\partial x(J,1)}$ | ...      | $\frac{\partial R(I,2n_c+1)}{\partial x(J,n_c)}$ | 0 ... 0                | $\frac{\partial R(I,2n_c+1)}{\partial x(J,2n_c+1)}$ | $\frac{\partial R(I,2n_c+1)}{\partial x(J,2n_c+2)}$ | $\frac{\partial R(I,2n_c+1)}{\partial x(J,2n_c+3)}$ | $\frac{\partial R(I,2n_c+1)}{\partial x(J,2n_c+4)}$ | 0        | $\frac{\partial R(I,2n_c+1)}{\partial x(J,2n_c+6)}$ |
| 0                                              | ...      | 0                                                | 0 ... 1                | 0                                                   | 0                                                   | 0                                                   | 0 1                                                 | 0        | 0                                                   |
| $\frac{\partial R(I,2n_c+3)}{\partial x(J,1)}$ | ...      | $\frac{\partial R(I,2n_c+3)}{\partial x(J,n_c)}$ | 0 ... 0                | $\frac{\partial R(I,2n_c+3)}{\partial x(J,2n_c+1)}$ | $\frac{\partial R(I,2n_c+3)}{\partial x(J,2n_c+2)}$ | $\frac{\partial R(I,2n_c+3)}{\partial x(J,2n_c+3)}$ | $\frac{\partial R(I,2n_c+3)}{\partial x(J,2n_c+4)}$ | 0        | $\frac{\partial R(I,2n_c+3)}{\partial x(J,2n_c+6)}$ |
| $\frac{\partial R(I,2n_c+4)}{\partial x(J,1)}$ | ...      | $\frac{\partial R(I,2n_c+4)}{\partial x(J,n_c)}$ | 0 ... 0                | $\frac{\partial R(I,2n_c+4)}{\partial x(J,2n_c+1)}$ | $\frac{\partial R(I,2n_c+4)}{\partial x(J,2n_c+2)}$ | $\frac{\partial R(I,2n_c+4)}{\partial x(J,2n_c+3)}$ | $\frac{\partial R(I,2n_c+4)}{\partial x(J,2n_c+4)}$ | 0        | $\frac{\partial R(I,2n_c+4)}{\partial x(J,2n_c+6)}$ |
| $\frac{\partial R(I,2n_c+5)}{\partial x(J,1)}$ | ...      | $\frac{\partial R(I,2n_c+5)}{\partial x(J,n_c)}$ | 0 ... 0                | $\frac{\partial R(I,2n_c+5)}{\partial x(J,2n_c+1)}$ | $\frac{\partial R(I,2n_c+5)}{\partial x(J,2n_c+2)}$ | $\frac{\partial R(I,2n_c+5)}{\partial x(J,2n_c+3)}$ | $\frac{\partial R(I,2n_c+5)}{\partial x(J,2n_c+4)}$ | 0        | $\frac{\partial R(I,2n_c+5)}{\partial x(J,2n_c+6)}$ |
| $\frac{\partial R(I,2n_c+6)}{\partial x(J,1)}$ | ...      | $\frac{\partial R(I,2n_c+6)}{\partial x(J,n_c)}$ | 0 ... 0                | $\frac{\partial R(I,2n_c+6)}{\partial x(J,2n_c+1)}$ | $\frac{\partial R(I,2n_c+6)}{\partial x(J,2n_c+2)}$ | $\frac{\partial R(I,2n_c+6)}{\partial x(J,2n_c+3)}$ | $\frac{\partial R(I,2n_c+6)}{\partial x(J,2n_c+4)}$ | 0        | $\frac{\partial R(I,2n_c+6)}{\partial x(J,2n_c+6)}$ |

$$J_{I,J} = \begin{bmatrix} \frac{\partial R(I,1)}{\partial x(J,1)} & \dots & \frac{\partial R(I,1)}{\partial x(J,n_c)} & 0 \dots 0 & \frac{\partial R(I,1)}{\partial x(J,2n_c+3)} & 0 & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots \vdots \vdots & \vdots & \vdots \vdots \vdots & & \\ \frac{\partial R(I,n_c)}{\partial x(J,1)} & \dots & \frac{\partial R(I,n_c)}{\partial x(J,n_c)} & 0 \dots 0 & \frac{\partial R(I,n_c)}{\partial x(J,2n_c+3)} & 0 & 0 & 0 \\ 0 & \dots & 0 & 1 \dots 0 & 0 & 0 & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots \vdots \vdots & \vdots & \vdots \vdots \vdots & & \\ 0 & \dots & 0 & 0 \dots 1 & 0 & 0 & 0 & 0 \\ 0 & \dots & 0 & 0 \dots 0 & 0 & 1 & 0 & 0 \\ \frac{\partial R(I,2n_c+4)}{\partial x(J,1)} & \dots & \frac{\partial R(I,2n_c+4)}{\partial x(J,n_c)} & 0 \dots 0 & \frac{\partial R(I,2n_c+4)}{\partial x(J,2n_c+3)} & 0 & 0 & 0 \\ 0 & \dots & 0 & 0 \dots 0 & 0 & 0 & 1 & 0 \\ 0 & \dots & 0 & 0 \dots 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

A similar procedure must be performed for oil-solid system. For oil-solid  $\ln K_i$  and  $L_g$  do not exist in the residual equations. The general structure of this system is as follows:

For oil – gas system  $L_s, N_{s2}, N_{s3}$  and  $K_s$  do not exist in the residual vector. A similar approach performs for building jacobian matrix of this system.

## 5. Results

In this section, the results of the implemented asphaltene model are presented for a simulation case. The results of the runs are compared against the CMG GEM results. In this work, the reservoir fluid used for studying asphaltene precipitation was obtained from Burke et al. (1990). **Table 2** shows the oil composition of reservoir fluid. Fluid characterization is the primary step in any compositional reservoir simulation activity. A PVT package is used for the characterization of the oil. **Table 3** shows that all the components are lumped into 15 components. The heaviest component of the oil ( $C_{31+}$ ) is divided into a non-precipitating component ( $C_{31A+}$ ) and a precipitating component ( $C_{31B+}$ ). The tuned EOS parameters used in this model is reported in **Table 4**. An oil reservoir has been constructed and related properties were assigned to investigate the deposition phenomena due to natural depletion (**Table 5**). Reservoir size and static reservoir properties such as absolute permeability and porosity were generated to study the deposition of asphaltene due to pressure depletion. The reservoir is one-dimensional homogenous with 500 feet in length, 10 feet in width and 10 feet in thickness. The porosity is 0.2 and permeability is 30md. The reservoir is discretized into 21 gridblocks. The reservoir is under primary recovery and there is only one production well in the middle of the reservoir. The producer operates under a constant bottomhole reservoir fluid rate (BHF) of 2.5 bbl/day. The parameters for the asphaltene precipitation model are given in **Table 6**. The input parameters for the deposition and flocculation models which is taken from Kohse and Nghiem (2004) are shown in **Table 7**.

**Figure 3** presents the pressure distribution along the length of the reservoir after 80 days of simulation time. Maximum pressure loss can be seen in the middle of the reservoir in gridblock(11,1,1) that the well is producing from. Also, we can see a symmetric distribution in this figure because of well location.

The results of precipitated and deposited asphaltene profile along the length of the reservoir after 80 days of simulation time are shown in Figure 4 and 5 respectively. There are less precipitation and deposition around the production well and it gradually increases as we go towards the other end of the reservoir.

Figure 6 presents pressure versus time for the grid block (11,1,1). As we can see from this figure, the initial reservoir pressure and saturation pressure are 6014.7 and 2950 psia, respectively. The saturation pressure will be reached after 30 days of production time. The pressure depletion continues to reach a value of 2223.4 at the end of simulation time.

The amount of precipitated asphaltene (Mass per Bulk volume  $\text{lb/ft}^3$ ) as a function of time during the simulation is shown in Figure 7. This figure presents that the precipitation occurs after 12 days of production time, reaches a maximum value around the saturation pressure and decreases as pressure drops further. Below the bubble point pressure ( $P_b$ ), the solubility of asphaltene in the liquid phase increases due to separation of the light components from oil and move into gas phases. Therefore, some precipitated asphaltene dissolve back into the oil solution.

The amount of flocculated asphaltene (Mass per Bulk volume  $\text{lb/ft}^3$ ) with respect to production time is illustrated in Figure 8. As can be seen, this figure has a similar trend to the precipitated asphaltene plot, but there is a time interval between the maximum value of precipitation and flocculation. This figure presents the amount of flocculated asphaltene reaches a maximum value at the 37 days of simulation and then it decreases towards a value of 0.03.

Figure 9 shows the amount of deposited asphaltene (Mass per Bulk volume  $\text{lb/ft}^3$ ) and rate of deposited asphaltene (Mass per Bulk volume per time  $\text{lb/ft}^3/\text{day}$ ) with respect to production time. This figure presents the buildup of deposited asphaltene in the well block. As can be seen, the asphaltene deposited per unit bulk volume of the reservoir gradually increases in

gridblock (11,1,1). This figure shows that rate of deposited asphaltene reaches a maximum value at the 35 days of simulation and then it decreases towards a value of less than 0.0002 (Mass per Bulk volume per time lb/ft<sup>3</sup>/day). This figure also presents that around the maximum value of flocculated asphaltene the rate of deposited asphaltene approaches a maximum value. As a result, the concentration of flocculated asphaltene plays an important role in the rate of deposited asphaltene.

The pressure, amount of precipitated, flocculated and deposited asphaltene as a function of time are shown in Figure 10. This figure can help for better understanding of asphaltene deposition process during natural depletion. As we can see from this figure, the maximum precipitation occurs around the saturation pressure of 2950 psia and the onset pressure of asphaltene is approximately 4500 psia. As can be seen, there is a time interval between the maximum value of precipitation and flocculation. Also, the reduction of maximum value in the precipitation is sharper than flocculation.

Figures 11 through 13 show the comparison of the precipitated asphaltene profile, flocculated asphaltene profile and the deposited asphaltene profile between Matlab code and CMG GEM during 80 days of simulation time, respectively.

## 6. Conclusion

A fully implicit compositional simulator is presented to model asphaltene deposition during natural depletion. A solid model is used to model asphaltene precipitation. A first-order chemical reaction is used to model forward and reverse rates for the conversion of precipitated asphaltene to flocculated asphaltene. Also, a deposition model including adsorption, pore throat plugging, and re-entrainment is used. The reduction in the rock porosity and permeability are also included in the asphaltene model. The governing equations for the asphaltene deposition modeling are presented followed by solution approach using the fully implicit scheme. The description of how to construct the elements of the Jacobian matrix

for modeling asphaltene precipitation and an explanation of how to overcome the problems that created in the especial cases in the structure of Jacobian matrix are presented. This paper provides a more detailed description and better understanding of complex modeling of asphaltene deposition during natural depletion.

## Nomenclature

$f_{s1}^*$  = reference solid fugacity (psi)

$f_{s1}$  = fugacity of solid  $s_1$  (psi)

$v_s$  = solid molar volume ( $\text{ft}^3/\text{lbmole}$ )

$f_{ig}$  = fugacity of component  $i$  in gas phase (psi)

$f_{io}$  = fugacity of component  $i$  in oil phase (psi)

$f_{n_c,o}$  = fugacity of asphaltene component in oil phase (psi)

$W_{\text{exp}}$  = Experimental results of weight percent of precipitated asphaltene

$W_{\text{model}}$  = Calculated results of weight percent of precipitated asphaltene

$N_i$  = moles of component  $i$  per pore volume ( $\text{lbmole}/\text{ft}^3$ )

$N_T$  = total number of moles per pore volume ( $\text{lbmole}/\text{ft}^3$ )

$N'_T$  = total number of moles without flocculated solid and deposited solid per pore volume ( $\text{lbmole}/\text{ft}^3$ )

$s_1$  = solid phase which is in equilibrium with oil and gas phases

$s_2$  = solid phase created from solid  $s_1$  via a chemical reaction

$s_3$  = solid phase created from solid  $s_2$  via a deposition equation

$N_{s1}$  = moles of solid  $s_1$  per pore volume (lbmole/ft<sup>3</sup>)

$N_{s2}$  = moles of solid  $s_2$  per pore volume (lbmole/ft<sup>3</sup>)

$N_{s3}$  = moles of deposited solid per pore volume (lbmole/ft<sup>3</sup>)

$Z_i$  = global mole fraction of component  $i$  in feed

$\phi_{ij}$  = fugacity coefficient of component  $i$  in phase  $j$

$K_i$  = equilibrium ratio of component  $i$

$K_{ig}$  = gas-oil equilibrium ratio of component  $i$

$K_{n_c s}$  = solid-oil equilibrium ratio for component  $n_c$

$K_0$  = original permeability before asphaltene precipitation (md)

$K$  = absolute or instantaneous permeability (md)

$K_{12}$  = forward rate of formation of solid  $S_2$  from solid  $S_1$  (1/day)

$K_{21}$  = reverse rate of formation of solid  $S_1$  from solid  $S_2$  (1/day)

$L_g$  = mole fraction of gas phase

$L_s$  = mole fraction of solid phase

$n_b$  = total number of gridcells

$n_c$  = number of hydrocarbon components

$A$  = gridblock cross-sectional area (ft<sup>2</sup>)

$C_{s_1 o}$  = concentration of suspended solid  $s_1$  in oil phase (lbmole/ft<sup>3</sup>)

$C_{s_2 o}$  = concentration of suspended solid  $s_2$  in oil phase (lbmole/ft<sup>3</sup>)

$C_{s_2^f}$  = volumetric concentration of flowing solid  $s_2$  per volume of oil

$u_o$  = oil phase Darcy velocity (ft/day)

$V$  = gridblock volume (ft<sup>3</sup>)

$T_k$  = transmissibility of phase k (k = o,g)

$K_{r_k}$  = relative permeability of phase k

$V_{cr,o}$  = critical oil phase interstitial velocity (ft/day)

$v_o$  = oil phase interstitial velocity, ft/day

$V_{s_2^d}$  = volume of deposited solid  $s_2$  per gridblock volume

$y_{ik}$  = mole fraction of component i in phase k (k = o,g)

$y_{sj}$  = mole fraction of suspended solid in oil phase (j = 1, 2)

$\alpha$  = surface deposition rate coefficient (day<sup>-1</sup>)

$\beta$  = entrainment rate coefficient (ft<sup>-1</sup>)

$\gamma$  = pore throat plugging rate coefficient (ft<sup>-1</sup>)

$\xi_k$  = molar density of phase k (k = o,g) (lbmole/ft<sup>3</sup>)

$\mu_k$  = viscosity of phase k (k = o,g) (cp)

$\gamma_k$  = gradient of phase k (k = o,g)

$\Delta l$  = gridblock distance (ft)

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Table 1 – Static precipitation test results as a function of pressure at 212 °F (from Burke et al. 1990)

| Pressure(psi) | Weight% of precipitated asphaltene ( $W_{Exp}$ ) |
|---------------|--------------------------------------------------|
| 1014.7        | 0.403                                            |
| 2950          | 1.037                                            |
| 3014.7        | 0.742                                            |
| 4014.7        | 0.402                                            |

**Table 2 - Composition of reservoir fluid**

| Component        | mole % | MW     |
|------------------|--------|--------|
| CO <sub>2</sub>  | 2.46   | 44.010 |
| N <sub>2</sub>   | 0.57   | 28.013 |
| C <sub>1</sub>   | 36.37  | 16.043 |
| C <sub>2</sub>   | 3.47   | 30.070 |
| C <sub>3</sub>   | 4.05   | 44.097 |
| i-C <sub>4</sub> | 0.59   | 58.124 |
| n-C <sub>4</sub> | 1.34   | 58.124 |
| i-C <sub>5</sub> | 0.74   | 72.151 |
| n-C <sub>5</sub> | 0.83   | 72.151 |
| FC <sub>6</sub>  | 1.62   | 86.000 |
| C <sub>7+</sub>  | 47.96  | 329    |

**Table 3 - Modeled fluid**

|                            |            |
|----------------------------|------------|
| Reservoir Temperature      | 212 °F     |
| Initial reservoir pressure | 6014.7 psi |

**composition**

| Component                        | mole % | MW      |
|----------------------------------|--------|---------|
| CO <sub>2</sub>                  | 2.46   | 44.010  |
| N <sub>2</sub>                   | 0.57   | 28.013  |
| C <sub>1</sub>                   | 36.37  | 16.043  |
| C <sub>2</sub>                   | 3.47   | 30.070  |
| C <sub>3</sub>                   | 4.05   | 44.097  |
| i-C <sub>4</sub>                 | 0.59   | 58.124  |
| n-C <sub>4</sub>                 | 1.34   | 58.124  |
| i-C <sub>5</sub>                 | 0.74   | 72.151  |
| n-C <sub>5</sub>                 | 0.83   | 72.151  |
| FC <sub>6</sub>                  | 1.62   | 86.000  |
| C <sub>7</sub> -C <sub>15</sub>  | 19.66  | 147.272 |
| C <sub>16</sub> -C <sub>25</sub> | 12.55  | 279.231 |
| C <sub>26</sub> -C <sub>30</sub> | 4.00   | 389.527 |
| C <sub>31A</sub> +               | 7.42   | 665.624 |
| C <sub>31B</sub> +               | 4.32   | 665.624 |

**Table 4 - EOS****parameters**

| Parameter | value     |
|-----------|-----------|
| $a$       | 437834.55 |
| $b$       | 3.66      |

**Table 5 - simulation input data**

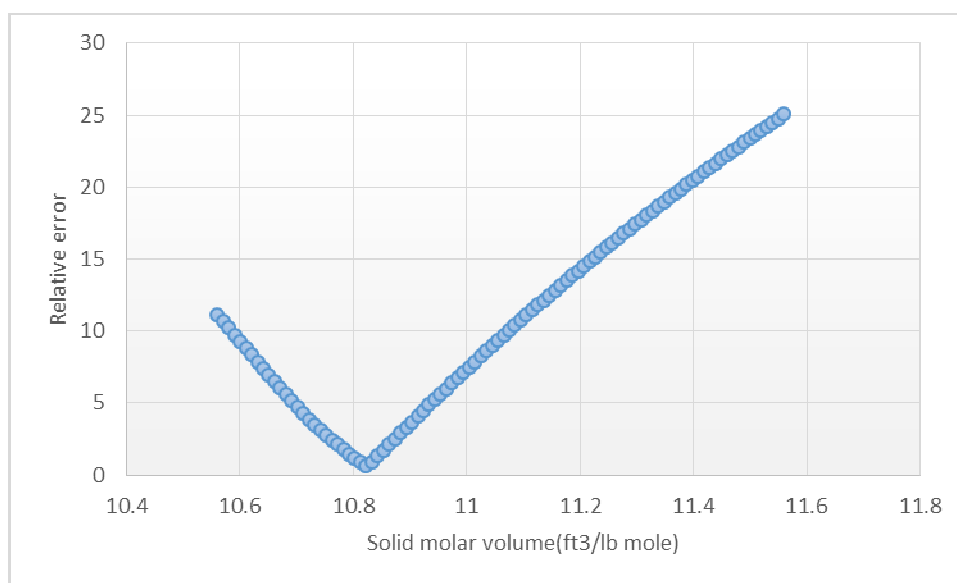
| Parameter                       | value                                |
|---------------------------------|--------------------------------------|
| No. of gridblocks               | 21                                   |
| Reservoir size                  | 500 *10*10 ft <sup>3</sup>           |
| Initial water saturation        | 0                                    |
| Porosity                        | 0.2                                  |
| Permeability                    | 30 md                                |
| Production constraint           | 2.5 bbl/day                          |
| Rock compressibility            | $5 \times 10^{-6}$ psi <sup>-1</sup> |
| Ref. P for rock compressibility | 14.7 psi                             |

**Table 6 - Asphaltene precipitation model parameters**

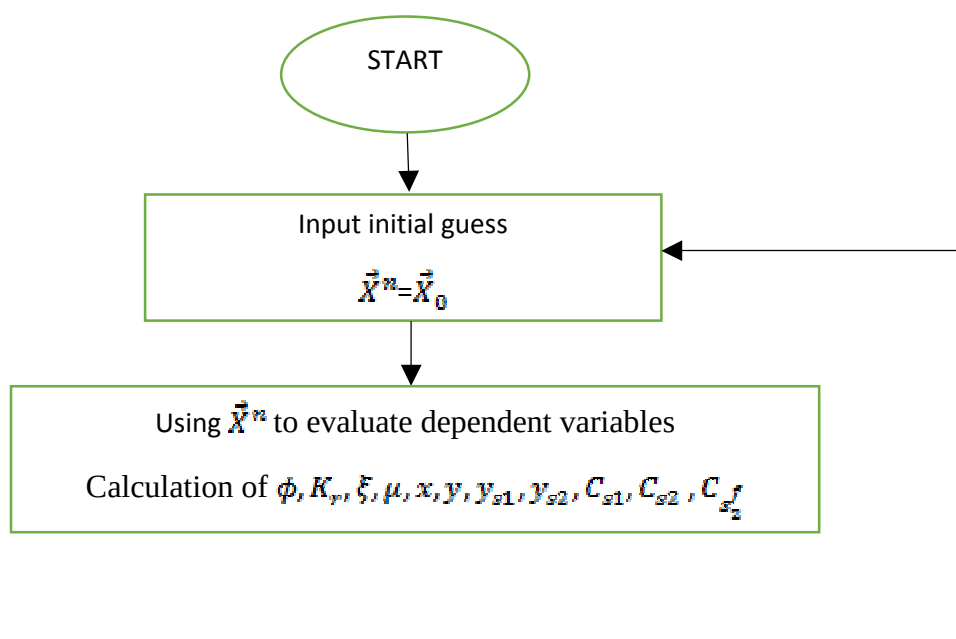
| Parameter         | value                            |
|-------------------|----------------------------------|
| $P^*$             | 4014.7 psi                       |
| $V_s$             | 10.81215 ft <sup>3</sup> /lbmole |
| $\ln f_{n_c s}^*$ | -25.223 psi                      |

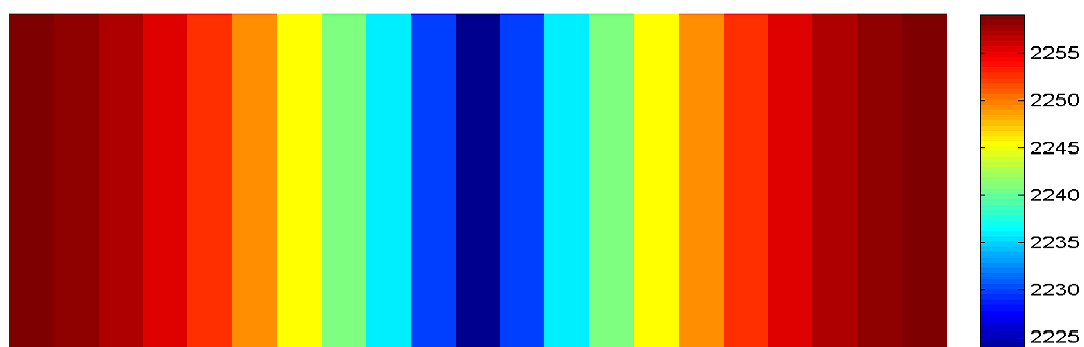
**Table 7 – Deposition and flocculation model parameters**

| Parameter                    | value |
|------------------------------|-------|
| k 12 (day -1 )               | 0.1   |
| k 21 (day -1 )               | 0.08  |
| $\alpha$ (day -1 )           | 0.01  |
| $\beta$ (ft <sup>-1</sup> )  | 0.0   |
| $\gamma$ (ft <sup>-1</sup> ) | 0.05  |
| $\sigma$                     | 150   |

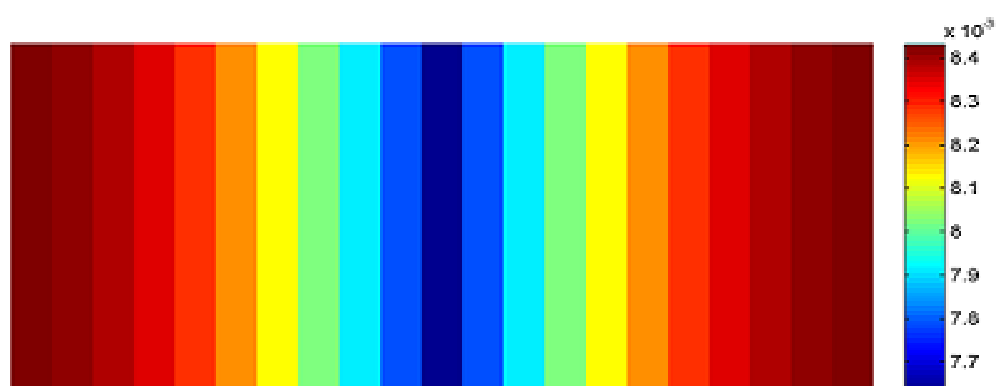


**Figure 1-** Relative error as a function of solid molar volume





**Figure 3** - Pressure profile after 80 days

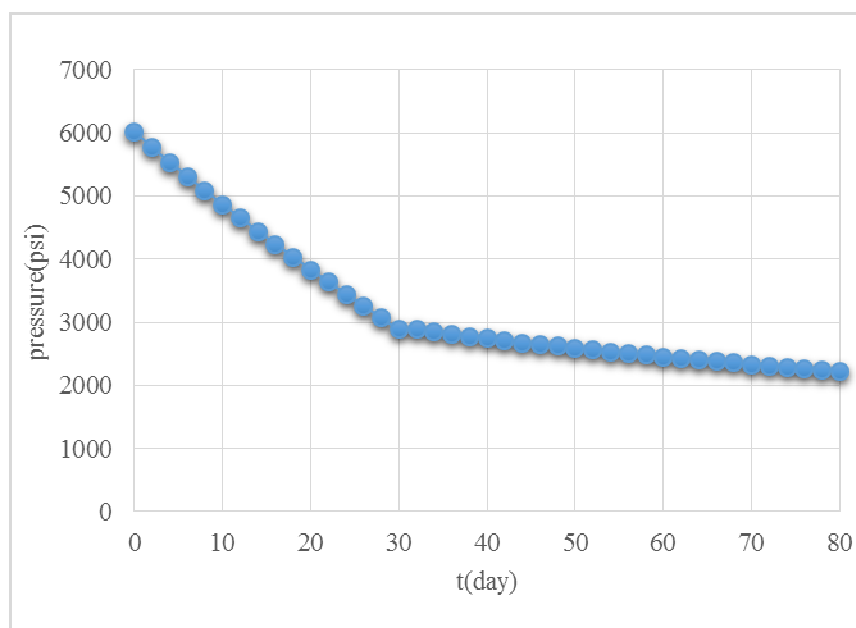


**Figure 4** - Asphaltene precipitated (Mass per Bulk Volume) along the reservoir after 80 days

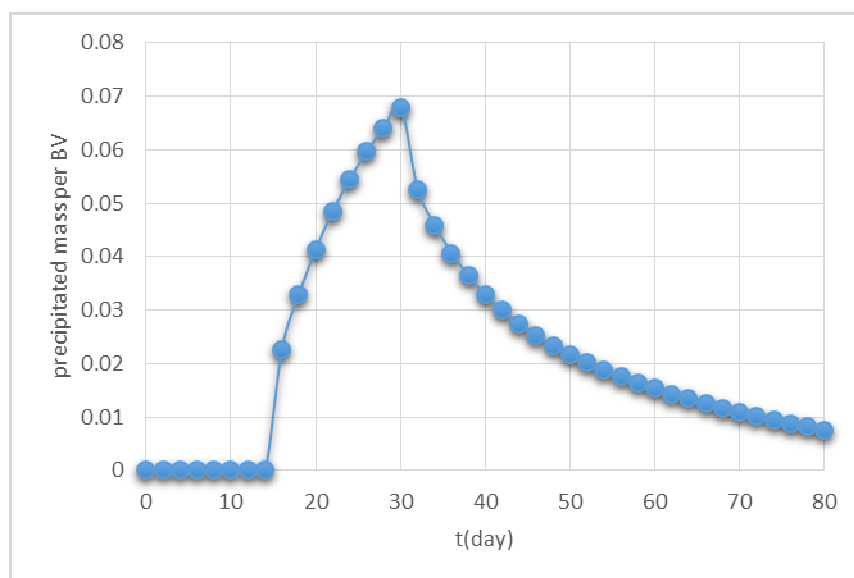


**Figure 5** - Asphaltene deposited (Mass per Bulk Volume) along the reservoir after 80 days

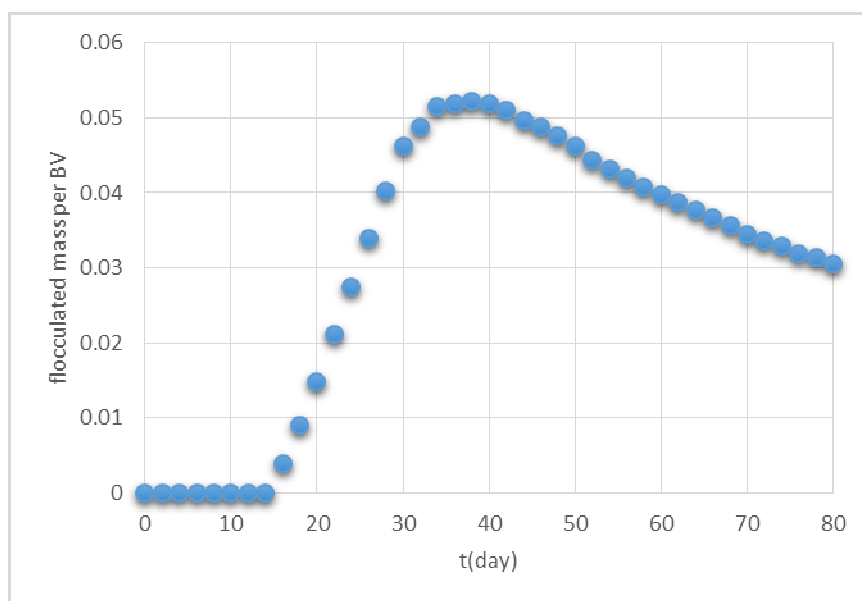




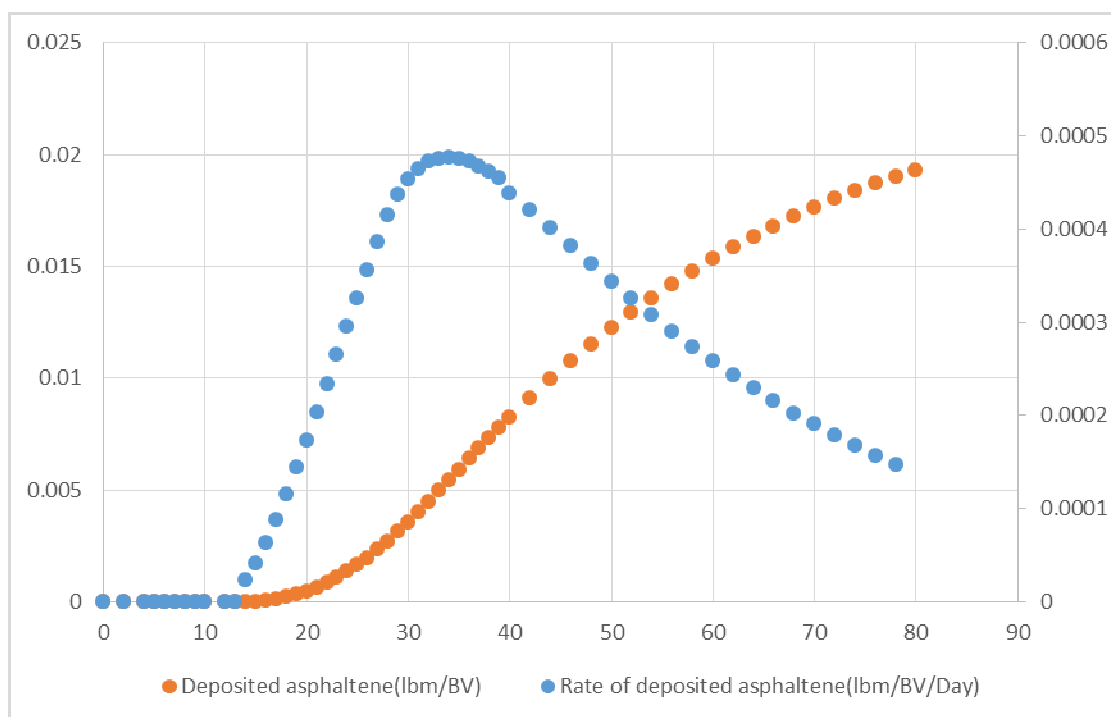
**Figure 6** – Pressure at gridblock(11,1,1)



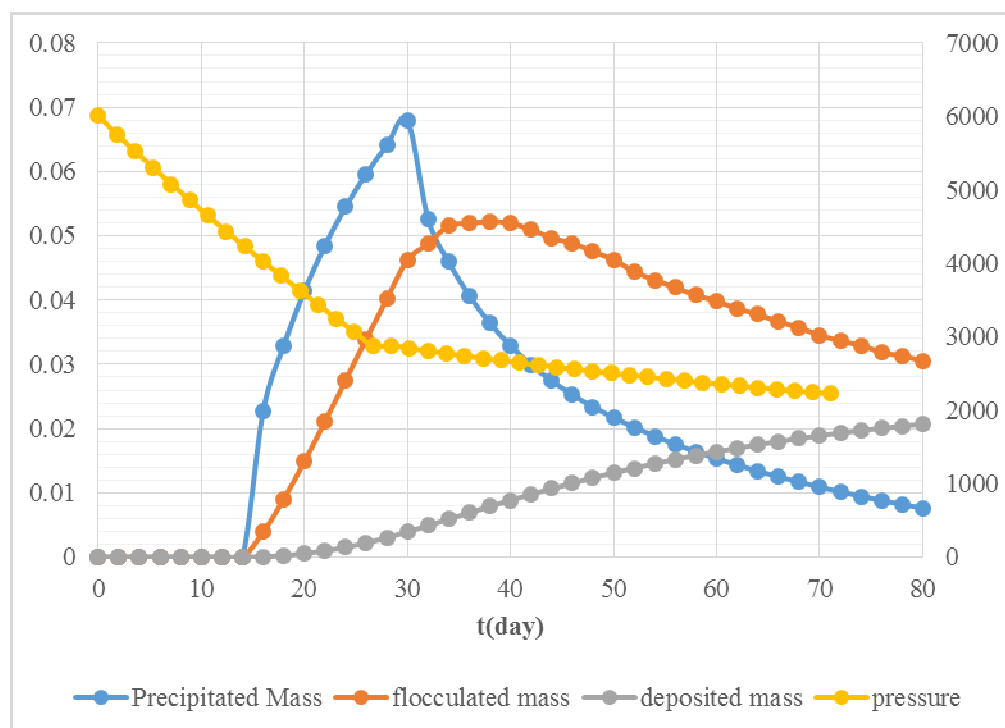
**Figure 7** – Asphaltene Precipitated Mass per Bulk Volume: (11,1,1)



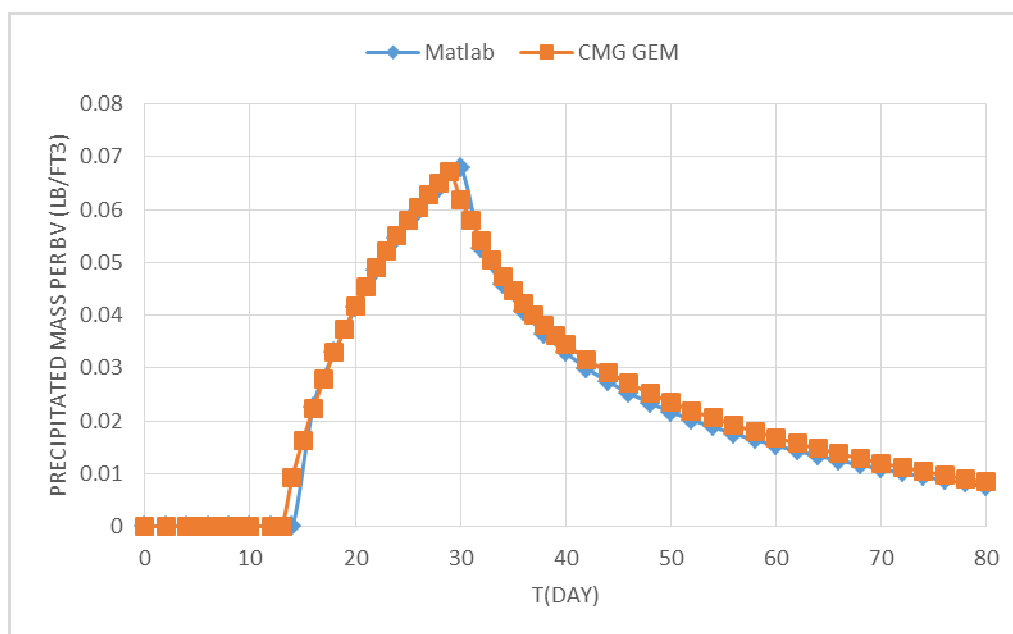
**Figure 8** - Asphaltene flocculated Mass per Bulk Volume: (11,1,1)



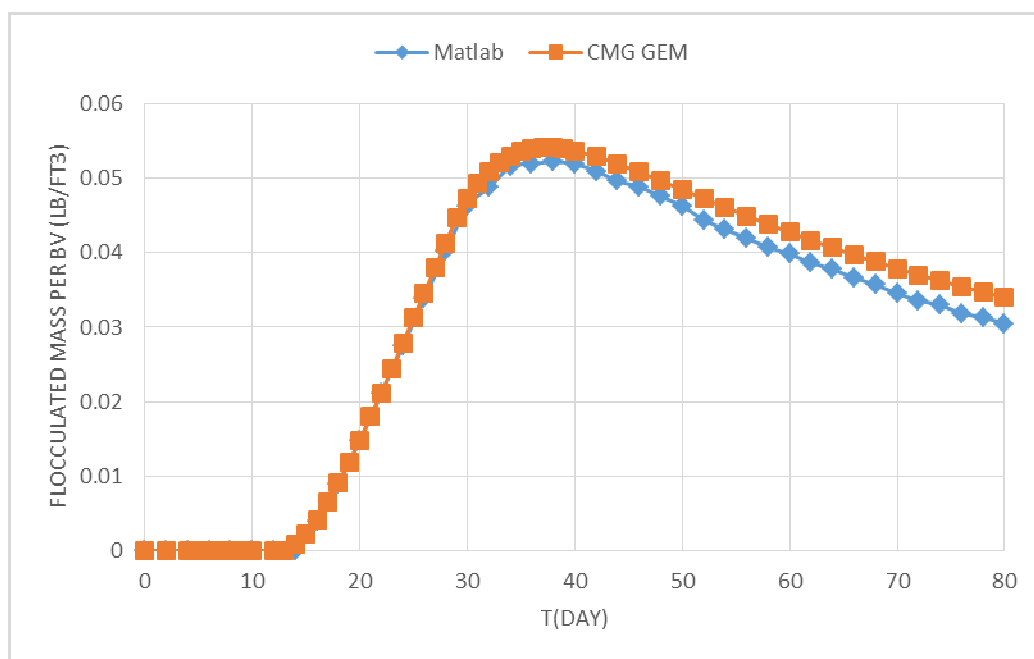
**Figure 9** - Deposited asphaltene and rate of deposited asphaltene: (11,1,1)



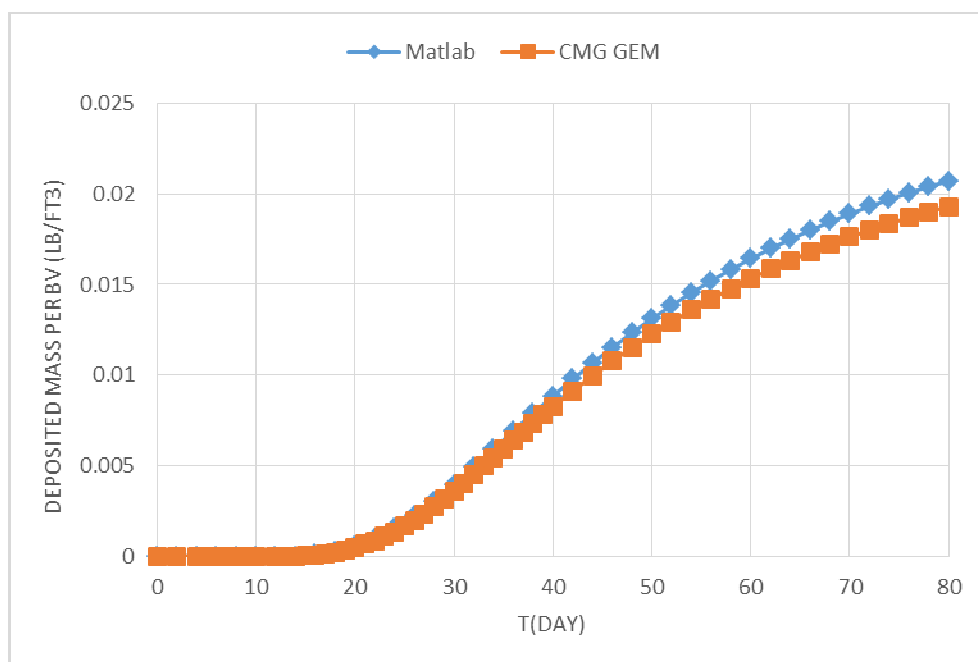
**Figure 10** - Precipitated mass per BV, flocculated mass per BV, deposited mass per BV and pressure



**Figure 11** - Asphaltene Precipitated Mass per Bulk Volume: (11,1,1)



**Figure 12** - Asphaltene Flocculated Mass per Bulk Volume: (11,1,1)



**Figure 13** - Asphaltene Deposited Mass per Bulk Volume: (11,1,1)